

Srijit Panja

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Education

Master's (M.Sc.) in Computer Science

Cochin University of Science and Technology

Specialization: Machine Intelligence

2019 - 2021

Institute: Indian Institute of Information Technology and Management-Kerala

Grade: First Class with Distinction

GPA: 8.65

Bachelor's (B.Sc.) with Math + Phy + C.S

University of Calcutta

Grade: First Division

2015 - 2019

Score: 62.08/100

Industry experience

Innovapptive

Sept 2024 - Present

Generative AI Lead

Hyderabad, India

- **Optimization of Existing AI Workflows for Factory Operations:** Led the enhancement of AI workflows for factory operations, significantly improving the efficiency and adaptability of AI systems tailored for frontline workers in varied roles. Introduced a two-step selection-retrieval process within the existing Retrieval-Augmented Generation (RAG) architecture, achieving a 1.5x boost in response accuracy. This was accomplished by devising a tailored retrieval mechanism that filters and narrows down candidate responses in two stages for precision. Engineered a custom prompt strategy using Selective Prompt Optimization (SPO) to handle inconsistencies and variability in user queries, enhancing contextual relevancy and response robustness. Redesigned the AI-driven dashboards workflow by transitioning from conventional schema-based indexing to schema-fields-definitions-based indexing, facilitating dynamic visual generation of daily plant operation metrics. Streamlined cost-efficiency by replacing proprietary, high-cost encoding models with performant, open-source encoders in the Equipment Issue Resolution (Work Order) workflow. Improved the workflow's throughput by restructuring sequential tasks into parallel processing units, employing the open-source Llama-3.1 and Llava-1.5 models as substitutes for expensive GPT-4 and GPT-4o instances, thereby reducing latency and operational costs.
- **Digitization of Manuals and Form Layouts:** Directed the technical initiatives for digitizing plant operation manuals and standardized forms such as maintenance logs, safety checklists, and troubleshooting records. Spearheaded the development of a two-flow form-JSON generation process optimized for efficiency and contextual adherence. The primary workflow was a RAG-based approach employing a custom-engineered prompt to produce nested, sectioned JSON outputs that align closely with client specifications. Leveraged an open-source Sentence Transformer encoder to create embeddings, which were stored and managed through Chroma as the vector database for fast retrieval and contextual alignment. Additionally, developed a sophisticated system capable of cloning existing paper-based forms by integrating an Image+Prompt input pipeline with a vision-language model (Vision-LLM). This process facilitated seamless transformation of analog documentation into digital replicas, streamlining data collection and accessibility in day-to-day operations.

Saras Analytics

June 2023 - Sept 2024

ML Engineer - Consulting

Hyderabad, India

- **Demand Forecasting for Health Supplement SKUs:** Built a Demand Forecasting Model for a US based Nutrition & Health Supplements brand. Identified categorization of SKU (Store

Keeping Units) depending on contribution to revenue & purchase type, and prioritized few SKUs. Correspondingly based on popular order types, executed a selective feature gathering. Devised a cohorts-based dataset (with avg. rebill % and cohort age as best features) for Subscriptions order type to train a non-linear regression model. Along same modelling approach, created an extraneous non-time series dataset for One Time Purchase orders. Final models post iterations with features and training patterns were used to deliver forecast for the next 4 months to aid stocking of SKUs. Task scheduling of the model with PostgreSQL connector into Python backend is under process.

- **E-commerce MMM Initiative:** Lead an internal initiative based on multi-client requirement to build in-house Media Mix Modelling capability for Marketing budget optimization for hit target LTV, Revenue & CAC KPIs. Procured Marketing spends data on campaign level and product level along with sales/customers/revenue information from BigQuery on various client accounts. Implemented statistical analysis to understand trend & correlation measures. Performed statistical tests to validate feature importances. Applied Autoregressive modelling on time series (considering adstocks) data and Bayesian optimization analysing several facets as Lag effect, Carryover effect, holidays, seasonality etc. Customized budget allocation on marketing channels for customers successfully delivering five pilot projects and user stories leading to conversion for paid engagements.
- **Daton Product Assistant (LLM Bot):** Lead development of a conversational chatbot for Saras's flagship product Daton, for user profiles - prospective customers and developers new to Daton to help assist in interacting with the product. Designed and implemented architecture to index Daton documentation on Pinecone VectorDB using Open Source Text embedding and retrieve context and generate response using RAG with Langchain framework, Llama-2 LLM being core to chat generation. MVP was deployed on Streamlit framework on a dedicated GCP instance. Daton assistant was attributed with Custom indexing and additionally Customer Service component with indexing of tickets from Freshdesk, Slack & ClickUp.
- **Daton Dashboards AI:** Developed a POC for automated Data sourcing from GCP BigQuery, Insights and Dashboards creation conversationally with Natural Language questions from client. Implemented a two-phase architecture on modified-RAG with schema context to VertexAI (Codey), generating SQL code for table filtration and PandasAI (with OpenAI GPT-3.5) for responses (as insights or visualization), UI being supported by Streamlit on local hosting. Alongside explored AWS Q capabilities for existing data connectors.

Cognida.ai

Data Scientist

October 2022 - June 2023

Hyderabad, India

- **LLM Bot project:** Co-led the creation of an advanced Natural Language Processing (NLP) chatbot for a US-based health insurance client, designed to enhance user interactions through contextual understanding and precise response generation. Utilized OpenAI's GPT-3.5 Chat Completion model as the foundation for generating conversational responses. Implemented a dual-database architecture using Qdrant for vector-based semantic search and CouchDB for document storage, enabling the system to efficiently index and retrieve client-specific data. Orchestrated the end-to-end development and deployment of APIs that handled document indexing, data retrieval, and advanced search operations, ensuring robust backend functionality. Collaborated closely with front-end development teams to guarantee seamless API integration and user experience consistency. Validated and tested API endpoints through SwaggerUI, achieving high compatibility and smooth data flow between client and server components.
- **EDEE Component:** ClientLed the development of a key API component within a comprehensive NLP-driven Question-and-Answer (QnA) pipeline, integrated into the Electronic Data Exchange Engine (EDEE) for an Ireland-based freight logistics company. This project involved leveraging Haystack's data annotation capabilities to prepare and manage datasets for fine-tuning a custom HuggingFace DistilBERT QnA model. Designed the backend processes to enhance data preprocessing, model training, and fine-tuning on client-specific datasets, thereby improving the model's accuracy and response relevance for real-world logistics applications. Delivered a tailored solu-

tion that significantly boosted the precision and contextuality of automated responses, addressing client-specific business needs effectively.

- **DocumentAI:** Developed a robust POC leveraging DocumentAI to automate the extraction of structured data from images and scanned PDF documents for a leading US-based engine equipment manufacturing company. The POC handled diverse document types, including invoices, billing statements, and packaging documentation, and parsed complex document structures using Tesseract OCR for initial text extraction and the LayoutLM model for advanced table recognition and layout understanding. Custom processing pipelines managed multi-page documents and complex table hierarchies, enabling the seamless extraction of nested data elements, with error-handling routines and data validation layers ensuring reliable, accurate outputs. Pre-processing techniques optimized document quality handling, such as skew correction, noise reduction, and format normalization. The POC was deployed on Heroku to demonstrate a scalable, end-to-end system integrated with data visualization tools for user-friendly data review and validation, effectively transforming unstructured inputs into structured, machine-readable formats, streamlining workflows, boosting operational efficiency, and reducing manual data entry errors, thus improving access to critical information and enhancing data utility for reporting and analytics.

Infogain

Software Engineer G2 - Data Science

August 2021 - October 2022

Bangalore, India

- **Patient No-Show prediction:** Developed an end-to-end machine learning solution to predict patient no-shows for a US healthcare provider, enhancing operational efficiency and patient management. Conducted extensive data cleaning and exploratory data analysis (EDA) on patient appointment data sourced from a PostgreSQL database, ensuring high data quality and feature consistency. Designed and standardized a feature selection framework aimed at optimizing model recall, achieving a performance benchmark of 0.80 recall. Fine-tuned ensemble models, specifically XGBoost and Random Forest classifiers, to balance accuracy and robustness for reliable predictions. Integrated a task-scheduling mechanism to automate the output of predictions, seamlessly updating the appointments table with real-time predictions and facilitating smooth adoption within the client's existing workflow.
- **Patient Risk Grouping:** Engineered a data-driven clustering solution for a chain of clinics in the US, focusing on the identification and categorization of high-risk patient segments. Employed a rigorous data preparation strategy that included data filtering and comprehensive feature-to-feature correlation analysis. Utilized hierarchical clustering (Dendrogram visualization) and K-Means algorithms to determine the most effective grouping strategy. The final clustering model demonstrated that over 75% of one identified cluster aligned with the clinic's known high-risk patient group. This innovative approach enabled more targeted patient care strategies and optimized resource allocation, providing actionable insights for improved clinical outcomes.
- **Healthcare data synthesis (as a Jr. Data Scientist):** Contributed to the development of a self-service Business Intelligence (BI) tool by generating synthetic text data to augment a limited text corpus for a US-based healthcare client. Leveraged a Variational Autoencoder (VAE) to produce realistic and contextually coherent synthetic data, enhancing the dataset's richness and representativeness. This synthetic data augmentation strengthened downstream analyses and machine learning model training, ultimately supporting the BI product's capability to deliver comprehensive insights and bolster informed decision-making. The initiative significantly improved data utility, enabling more robust model development and better business outcomes.

Teaching experience

1StepGrow Academy	Data Science Trainer	Oct, 2023 - Present
• <u>Data Science course:</u>	Teaching an end-to-end online Data Science course of 3 months with Python, Stats, ML & DL for working professionals for transition to Data Science domain.	
Digital Edify	Data Science Trainer	May, 2024 - Sep, 2024
• <u>Data Science course:</u>	Teaching a detailed Data Science (with Generative AI) course of 4 months with Python, Stats, ML, DL, NLP, Time-Series, Computer Vision & Gen AI for undergrad students & working professionals.	
MachIT	Data Science Faculty	Sep, 2023 - May, 2024
• <u>Data Science sessions:</u>	Delivered 2-month online courses for college students and recent grads to get introduced to Python, Stats & ML for Data Science career.	
ExcelR Solutions	Data Science Trainer	Jan, 2023 - May, 2024
• <u>Data Science sessions:</u>	Taught end-to-end Data Science courses of 3 months both online and offline to undergrads, recent grads, and working professionals along with Interview Preparation sessions for entry/switch to Data Science.	
Masai School	Data Science Instructor	Sep, 2022 - Dec, 2022
• <u>Prob & Stats classes:</u>	Hosted online classes for Probability and Statistics module under Data Analytics (DA) track.	

Research & Publications

S.Panja, "Information Retrieval Systems in Healthcare: Understanding Medical Data Through Text Analysis", in *Book: Transformative Approaches to Patient Literacy and Healthcare Innovation, Chapter 9: IGI Global, Feb 2024* DOI: [10.4018/979-8-3693-3661-8.ch009](https://doi.org/10.4018/979-8-3693-3661-8.ch009), ISBN: 9798369336618

Srijit Panja. "Gravity Score: New Metric to Measure Plagiarism in Text Documents using the Concept of Gravitational Force." *Journal of Mechanics of Continua and Mathematical Sciences*, Vol.-17, No.-12, December (2022) pp 14-30 DOI: [10.26782/jmcms.2022.12.00002](https://doi.org/10.26782/jmcms.2022.12.00002)

Panja, S., and A. P. James. "Belief Index for Fake COVID19 Text Detection." *2020 IEEE Recent Advances in Intelligent Computational Systems (RAICS)*. IEEE, 2020. DOI: [10.1109/RAICS51191.2020.9332508](https://doi.org/10.1109/RAICS51191.2020.9332508)

Panja, S., Akshay Kumar Maan, and Alex Pappachen James. "Vilokana-lightweight COVID19 document analysis." *2020 IEEE 63rd International Midwest Symposium on Circuits and Systems (MWSCAS)*. IEEE, 2020. DOI: [10.1109/MWSCAS48704.2020.9184598](https://doi.org/10.1109/MWSCAS48704.2020.9184598)

Technical skills

Programming Languages/Tools

C, Java, Python, R, SQL, Unix, Machine Learning, Deep Learning, Data Science, NLP, Prompt Engineering, G Co-lab, Spyder, Jupyter Notebook, VSCode, Numpy, Scipy, Pandas, Scikit-Learn, Keras, Tensorflow, NLTK, Spacy, RNN, LSTM, Word2Vec, BERT, Transformers, RoBERTa, DistilBERT, Matplotlib, Seaborn, Shap, PowerBI, Streamlit, Docker, CouchDB, QDrant, OpenAI GPT models

Representing India at UN

Volunteer Data Scientist, United Nations Development Programme - SDG AI Lab, IICPSD, Istanbul, Turkey (Remote - July to Sept, 2021)

Open Source contribution

- **AI Poet (on Github)** at **Global LLM Streamlit Hackathon, September'23** - [press release at IssueWire](#)
- **AI can now sit under a tree and write poems – new Indian AI app, AIPoet heightens innovation** - [press release at PRLog](#)

Media presence

- **Hindustan Times, The Hindu, The New Indian Express, ANI, Economic Times** - work on Indian Covid Search Engine, Vilokana (with IIITM-K) : [References of this Wiki Article](#)
- **Technology Feature, Nature Journal** - work on Indian Covid Search Engine, Vilokana : [Artificial-intelligence tools aim to tame the coronavirus literature](#)
- **Podcast with CollectiWise** - Exploring the Relationship between Consciousness and AI : [YouTube](#) / [Audio](#)

Past internships

Nggawe Nirman Technologies	Machine Learning Intern, Feb - July, 2021
ValueMantra	AI and Data Management Intern, Jan - Mar, 2021
IIITM-Kerala	Research Intern, Apr - Dec, 2020
B-Aegis Life Sciences & Research	AI Intern, May - Jun, 2020
Clootrack	Data Science Intern, Dec, 2019 - Mar, 2020

I write about Data Science on

<https://medium.com/@srijitpanja>